

Results Pack

Compression of radar range-FFT data on Kintex UltraScale+ — all measured results, updated 24 September 2026 with placed-and-routed hardware

Jordan Mohindra (z5418455) · Dataset: ColoRadar cascade, 50 frames · Device: xcku5p-ffvb676-2-e @ 100 MHz

Reading guide. Four designs appear throughout: **DRHE-1** and **LPC-1,2** are the algorithms as originally implemented (predicting from the immediately preceding ramp); **DRHE- n** and **LPC- $n,2n$** are the TDM-corrected versions (predicting from the preceding ramp of the same transmitter, $n = n_{\text{Tx}} = 12$).

1 Phase 1 — MATLAB reference benchmark

Table 1: Single frame (**frame_0**), 16 RX, 190 reference detections.

Algorithm	CR	FN (%)	FDR (%)	NF (dBFS)	SNR (dB)
FP16	1.00	0.00	0.00	−105.858	36.256
DRHEfp16	1.05	0.00	0.00	−105.858	36.256
FX16	1.00	3.68	0.00	−105.591	36.150
DRHE	3.32	3.68	0.00	−105.591	36.150
BAQ	3.76	4.74	3.72	−106.039	36.562
RDVLE	1.12	20.53	2.58	−102.476	33.791
RDVLE_UDRE	2.56	26.32	2.78	−102.002	33.663
LPC	3.39	3.68	0.00	−105.591	36.150

Table 2: 50-frame average, 16 RX, 9,934 reference detections. *This is the headline benchmark.* FX16, DRHE and LPC share detection metrics by construction — all three reconstruct the quantised data exactly.

Algorithm	CR	FN (%)	FDR (%)	NF (dBFS)	SNR (dB)
FP16	1.00	0.00	0.00	−105.638	35.504
DRHEfp16	1.04	0.05	0.01	−105.637	35.505
FX16	1.00	3.77	1.11	−105.374	35.359
DRHE	3.30	3.77	1.11	−105.374	35.359
BAQ	3.76	10.44	9.69	−105.080	35.081
RDVLE	1.12	24.01	4.62	−102.275	33.165
RDVLE_UDRE	2.56	28.73	4.44	−101.782	33.004
LPC	3.37	3.77	1.11	−105.374	35.359

2 Phase 2 — bit-exact verification in hardware simulation

Table 3: Vitis HLS C-simulation, all 50 frames, 4 RX, 4,915,200 complex samples. **Every design reconstructs every sample exactly.**

Design	Mean CR	MaxDiff	MSE	Frames lossless
DRHE-1	3.30987	0	0	50 / 50
DRHE-n	3.75082	0	0	50 / 50
LPC-1,2	3.38497	0	0	50 / 50
LPC-$n,2n$	3.69681	0	0	50 / 50

Table 4: Agreement between MATLAB and hardware. For LPC the entire residual difference is AXI packet padding — the underlying bit counts are identical.

Quantity	Value
LPC: MATLAB, double-precision coefficients	3.38541
LPC: MATLAB, 16-bit quantised coefficients	3.38541
cost of coefficient quantisation	0.0000 %
LPC: MATLAB, quantised + 256-bit AXI padding	3.38497
LPC: Vitis HLS measured	3.38497
frames where the padding model matches	50 / 50 (to 4.9×10^{-6})
worst single frame	0.00091 (tolerance ± 0.01)
DRHE: hardware bits above MATLAB, per frame	15–252 (mean 134)
one AXI packet	256 bits

3 Phase 2.4 — synthetic edge cases

Table 5: DRHE, 8 synthetic cases. The first run exposed the -32768 defect (3 failures); after the saturation + closed-loop fix all 8 pass. The **random-int16** case is the proof the closed loop works — the clamp fires repeatedly across all 192 ramps and the error still never exceeds 1 LSB.

Case	Input	CR	MaxDiff		Result
			before	after	
all-zeros	every sample 0	4.0000	0	0	PASS
dc-constant	Re=1000, Im= -500 everywhere	3.5867	0	0	PASS
random-int16	uniform random int16	0.6037	60475	1	PASS
tone-192ramp	correlated tone, 192 ramps	3.2777	0	0	PASS
tone-1ramp	same tone, 1 ramp	0.6214	0	0	PASS
max-amplitude	all $+32767$	3.0132	0	0	PASS
max-alternating	$+32767 / -32768$ per ramp	3.0326	65535	1	PASS
int16-min	all -32768	3.0371	65535	1	PASS

Table 6: LPC, same 8 cases, all pass. LPC’s ratios are lower on the synthetic cases because the 32,768-bit coefficient table is charged whatever the data does — on **all-zeros** the residual bits are identical and the 4.00 vs 3.84 gap is exactly that overhead.

Case	CR	MaxDiff	Result
all-zeros	3.8400	0	PASS
dc-constant	3.3758	0	PASS
random-int16	0.6000	0	PASS
tone-192ramp	1.1006	0	PASS
tone-1ramp	0.2771	0	PASS
max-amplitude	0.9868	0	PASS
max-alternating	0.9868	1	PASS
int16-min	0.9868	1	PASS

Note on random-int16. $CR < 1$ is correct behaviour, not a fault: a Huffman/append coder cannot leave incompressible data unchanged, it expands it. A deployment needs a raw-passthrough fallback if such data is possible.

4 Phase 3 — synthesis, implementation and RTL co-simulation

Table 7: Resources and timing of all four designs, placed and routed (Vivado 2026.1, out of context) on the **xcku5p-ffvb676-2-e** at 10 ns. Device capacity: 216,960 LUT, 433,920 FF, 960 BRAM_18K, 1,824 DSP. Frame latency is measured in RTL co-simulation. Kiem’s figures are post-route on an **XCZU3CG-SFVC784-1-e**; his BRAM is 10 36 kbit tiles, shown as 20 BRAM_18K. The HLS estimates used in earlier versions of this pack are in Table 8.

	DRHE-1	DRHE- n	LPC-1,2	LPC- $n,2n$	Kiem
LUT	35,343	34,804	29,772	34,503	45,892
FF	23,546	23,514	13,601	20,186	9,243
BRAM_18K	40	80	244	216	20
DSP	132	132	168	158	44
Routed $F_{\max}$ (MHz)	135.5	131.0	118.6	106.3	≥ 100
WNS at 10 ns	2.621	2.367	1.569	0.595	—
Dynamic power (W)	0.562	0.657	0.709	0.555	—
Worst loop II	2	2	3	1	1
Pipeline depth	58	58	—	—	23
Frame latency (cyc)	61,253	61,253	197,277	325,166	—
Frame buffer	none	none	3.0 Mibit	3.0 Mibit	—

Table 8: HLS estimates against the same designs after Vivado placement and routing, **xcku5p-ffvb676-2-e**, 10 ns constraint. BRAM in BRAM_18K units. $F_{\max}$ is $1/(10 \text{ ns} - \text{WNS})$ after routing, and HLS’s own estimate before it. Every figure is read from the tool reports by `collect_impl.py`.

Design	LUT		FF		DSP		BRAM_18K		$F_{\max}$ (MHz)	
	HLS	impl.	HLS	impl.	HLS	impl.	HLS	impl.	HLS	impl.
DRHE-1	115,534	35,343	57,105	23,546	132	132	40	40	81.9	135.5
DRHE- n	115,736	34,804	57,263	23,514	132	132	80	80	81.9	131.0
LPC-1,2	80,528	29,772	17,393	13,601	168	168	256	244	85.4	118.6
LPC- $n,2n$	86,298	34,503	23,154	20,186	156	158	192	216	85.4	106.3

Table 9: Throughput. **Two corrections:** Kiem’s published 11.92 Gbit/s is in *gibibits* ($11.92 \times 2^{30}/10^8 = 127.99$ bits/cycle, decimal 12.80 Gbit/s), and it is *theoretical* — bus width $\times$ clock — not measured. Compare theoretical with theoretical (a factor of 2, all from II).

	DRHE	LPC	Kiem
Worst loop II	2	3	1
Theoretical @ 100 MHz ($128f/II$)	6.40 Gbit/s	4.27 Gbit/s	12.80 Gbit/s
Measured @ 100 MHz (co-sim)	5.14 Gbit/s	1.60 Gbit/s	not reported
Measured bits per cycle	51.4	15.9	—
Throughput @ routed $F_{\max}$	6.96 Gbit/s	1.89 Gbit/s	—
IP pipeline latency	580 ns	—	230 ns

Table 10: What the numbers are judged against (report Chapter 7). ColoRadar: 128 bins, 192 ramps, ramp interval 78.5 μ s, frame acquisition 15.07 ms, 5 Hz. Kiem’s sensor: 512 bins, 1024 ramps, $\approx 20.5 \mu$ s ramps, 15 ms compression budget. LPC figures are for lags n_{Tx} , $2n_{Tx}$.

Yardstick	DRHE- n	LPC- $n, 2n$
Required rate, 4 RX (0.209 Gbit/s) — headroom @ 100 MHz / routed $F_{\max}$	24.6 $\times$ / 32.2 $\times$	4.6 $\times$ / 4.9 $\times$
Required rate, 16 RX (0.835 Gbit/s) — headroom	6.2 $\times$ / 8.1 $\times$	1.16 $\times$ / 1.23 $\times$
Kiem’s sensor rate (3.2 Gbit/s) — headroom	1.6 $\times$ / 2.1 $\times$	—
Cycles per ramp used / available (7,850)	319 (4.1%)	ingest 128
Added latency after last ramp @ 100 MHz	3.2 μ s	3.01 ms
Added latency scaled to Kiem’s frame (budget 15 ms)	$\approx 11 \mu$ s	≈ 53 ms (fails)
Frame buffer scaled to Kiem’s frame	none	67 Mbit (exceeds on-chip)
LUT as % of Kiem’s XCZU3CG (70,560), routed	49%	49%
LUT / FF / BRAM / DSP vs Kiem, routed	0.76 / 2.54 / 4.0 / 3.0 $\times$	0.75 / 2.18 / 10.8 / 3.6 $\times$
BRAM used vs theoretical minimum	80 vs 53	192 vs 171
CR vs no-prediction baseline (3.489)	+7.5%	+6.0%

Table 11: LPC-1,2 cycle breakdown. The model is accurate to 0.08 %. For LPC- $n, 2n$ the same model predicts 246,272 against 325,166 measured — the 32 % gap is pipeline drain from 512 short loop invocations ($512 \times (71+56) \approx 65,000$ cycles).

Stage	Iterations	II	Cycles
Ingest	24,576	3	73,728
Solve + residual	98,304	1	98,304
Emit coefficients	512	1	512
Emit residuals	24,576	1	24,576
Model total			197,120
Measured (co-simulation)			197,277

4b Accounting for every difference from Kiem (report Chapter 7)

Kiem’s design was rebuilt from his thesis (Section 4.3.4: his fixed-point types and functions, 58-bit packer, open loop, $II = 1$) in the same tool, for the same device, and joined to DRHE-1 by a ladder of single-change variants. Every row below is a lossless compressor (C-simulation on ColoRadar; step 3 also in RTL co-simulation), placed and routed.

Table 12: The rebuilt design against Kiem’s published compression IP (his Table 5.3 and Section 5.2.3). Replica figures are post-route, out of context, on the **xcku5p** at speed grades -2 and -1 ; Kiem’s are post-route, in system, on the **XCZU3CG-SFVC784-1-e**. Kiem’s BRAM is given in 36 Kb tiles and converted.

	Replica (-2)	Replica (-1)	Kiem [Kiem 2025]
LUT	17,141	17,228	45,892
FF	5,050	5,598	9,243
DSP	48	48	44
BRAM (36 Kb tiles)	10	10	10
Initiation interval	1	1	1
Pipeline depth (HLS, cycles)	12	14	23
Achieved clock (MHz)	129.5	125.0	100 (met)
Compression ratio (ColoRadar, 10 frames)	3.31904		—

Table 13: From DRHE-1 to the replica of Kiem’s design, one design decision per step. Every row is a lossless compressor implemented through placement and routing on the `xcku5p-ffvb676-2-e` at 10 ns. II and depth are from HLS; everything else is post-route. BRAM in BRAM_18K units.

Step	Change from the row above	LUT	FF	DSP	BRAM	II	Depth	MHz
0	DRHE-1 as built	35,343	23,546	132	40	2	58	135.5
1	packer shift offset made unsigned	33,502	24,423	132	40	2	58	128.4
2	Kiem’s 58-bit per-channel packer	31,124	21,261	127	40	2	62	120.6
3	state cleared by read, not write	52,432	30,373	264	40	1	59	131.4
4	Kiem’s fixed-point arithmetic	16,882	5,380	48	20	1	16	133.3
5	open loop = replica of Kiem	17,141	5,050	48	20	1	12	129.5
—	Kiem, published [Kiem 2025]	45,892	9,243	44	20	1	23	100

Table 14: Side experiments. “Fixed-point DRHE-1” changes only the arithmetic of DRHE-1. The replica variants test explanations for the gap between the replica and Kiem’s published LUT and FF counts.

Variant	LUT	FF	DSP	BRAM_18K	Depth	MHz
DRHE-1 (control)	35,343	23,546	132	40	58	135.5
Fixed-point DRHE-1 (II 2, closed loop)	20,549	9,230	44	20	12	102.8
Replica	17,141	5,050	48	20	12	129.5
Replica, speed grade −1	17,228	5,598	48	20	14	125.0
Replica, literal packer	18,576	6,214	46	20	12	134.8
Replica, 23-cycle pipeline	17,060	5,217	48	20	23	132.2
Kiem, published	45,892	9,243	44	20	23	≥100

Reading. Block RAM 2×: 32-bit state words (the replica matches Kiem exactly). DSP 3×: every DSP is a float operator; 6× at II = 1 (step 3, 264). II 2: one extra state write per cycle (step 3 reaches II 1; measured throughput 8.76 Gbit/s against 5.14). Depth 58: 26 cycles of closed loop plus float operator latency. LUT and FF: float operators are 69% of DRHE-1’s LUTs and 54% of its FFs; the packer another 4,200 LUTs. The replica against Kiem’s own build (2.68× LUT, 1.83× FF): speed grade, literal packer and pipeline depth together explain about 45% of the FF gap and 5% of the LUT gap; the rest is not a property of the design his thesis describes.

5 The audit — what the compression ratio measures

Table 15: The no-prediction baseline. Raw FX16 values coded through the *identical* S4 + Huffman back end with no predictor. 5 frames, 4 RX. Both predictors as implemented perform *worse* than doing nothing.

Configuration	bits/value	CR	vs. no prediction
No prediction (entropy coder only)	4.586	3.489	—
DRHE-1 (as implemented)	4.821	3.319	0.951×
LPC-1,2 (as implemented)	4.710	3.397	0.974×

Table 16: Entropy check, independent of the coder. The predictor was *increasing* the information content of the data.

Quantity	Value
Order-0 entropy, raw FX16 codes	4.094 bits
Order-0 entropy, DRHE-1 residuals	4.538 bits
Achieved bits/value, no prediction	4.586 bits
Huffman overhead above entropy	0.283 bits (6.2%)

Table 17: Decomposition of the compression ratio. Most of it is unused fixed-point container, not algorithm.

Source	Factor	Evidence
Unused container headroom	× 1.63	peak raw ADC sample 914 of 32,767; peak FX16 code 436 ⇒ 9.8 of 16 bits used
Spectral sparsity	× 2.14	9.8 → 4.59 bits, genuine
Subtotal, no prediction	× 3.489	
Prediction, as implemented	× 0.95	negative contribution
Prediction, corrected	× 1.08	

Table 18: Prediction axis study. 5 frames, 4 RX, 128 bins, 192 ramps. Gain is relative to no prediction; below 1.000 means the predictor is doing harm. Row 5 is the obvious alternative that fails: per-transmitter blocks multiply the coefficient overhead by 12 (1.04 % $\rightarrow$ 12.5 % of the frame) and shorten each fit from 192 samples to 16.

Configuration	bits/value	CR	gain
No prediction	4.586	3.489	1.000
DRHE, lag 1 (<i>as implemented</i>)	4.821	3.319	0.951
DRHE, lag n_{Tx}	4.240	3.774	1.082
LPC, lags 1, 2 (<i>as implemented</i>)	4.710	3.397	0.974
LPC, per-transmitter blocks, $N = 16$	6.093	2.626	0.753
LPC, lags $n_{Tx}, 2n_{Tx}$	4.269	3.748	1.074

Table 19: Headline before/after, measured in hardware over all 50 frames. Still exactly lossless in both corrected designs.

Design	CR	MaxDiff	Δ CR	Cost of the correction
DRHE-1	3.30987	0	—	—
DRHE-n	3.75082	0	+13.3%	+40 BRAM (4% of device); logic, F_{max} , latency unchanged
LPC-1,2	3.38497	0	—	—
LPC-$n, 2n$	3.69681	0	+9.2%	−64 BRAM, II 3 $\rightarrow$ 1, but $1.65\times$ latency

6 Where the bits go

Table 20: Bit budget by S4 category, one frame, 4 RX, 196,608 values. “% values” is how often a category occurs; “% bits” is how much of the output it consumes. Under DRHE-1, categories 5–8 are 5.53 % of values but 13.84 % of bits.

S_4	cost	No prediction		DRHE-1		DRHE- n	
		% values	% bits	% values	% bits	% values	% bits
0	4	16.71	14.61	11.78	9.81	18.85	17.87
1	4	28.57	24.97	22.01	18.33	32.94	31.22
2	4	31.45	27.49	31.95	26.60	34.40	32.60
3	5	14.66	16.02	22.21	23.12	11.31	13.40
4	6	4.31	5.65	6.46	8.07	1.49	2.12
5	10	1.75	3.82	2.36	4.92	0.50	1.18
6	12	1.15	3.01	1.43	3.58	0.29	0.82
7	14	1.14	3.47	1.09	3.17	0.18	0.59
8	16	0.19	0.66	0.65	2.17	0.03	0.13
9	18	0.07	0.29	0.06	0.23	0.01	0.05
bits/value		4.576		4.803		4.220	
CR		3.497		3.331		3.792	

Table 21: Residual magnitude distribution, 5 frames, 4 RX. LPC figures exclude the coefficient table.

Configuration	median $ d $	99th pct $ d $	bits/value
No prediction	2.0	80.6	4.586
DRHE, lag 1	2.2	102.4	4.821
DRHE, lag n_{Tx}	1.6	15.8	4.240
LPC, lags 1, 2	2.0	63.4	4.543 [†]
LPC, lags $n_{Tx}, 2n_{Tx}$	1.0	8.2	4.102[†]

[†] residual bits only; add 0.167 bits/value for the coefficient table.

Table 22: Worked example. Sixteen consecutive ramps of range bin 75, channel 1, frame 0, traced through the *same* DRHE predictor at two lags. Real part; the imaginary part behaves identically. Only the lag differs.

m	TX	x	from ramp $m - 1$				from ramp $m - n_{\text{Tx}}$			
			$\hat{x}$	d	S_4	bits	$\hat{x}$	d	S_4	bits
168	0	-69	67	-136	8	16	-72	3	2	4
169	1	-52	-75	23	5	10	-49	-3	2	4
170	2	59	81	-22	5	10	59	0	0	4
171	3	-6	64	-70	7	14	-5	-1	1	4
172	4	-29	24	-53	6	12	-30	1	1	4
173	5	-15	-29	14	4	6	-18	3	2	4
174	6	59	-6	65	7	14	63	-4	3	5
175	7	-20	-83	63	6	12	-23	3	2	4
176	8	-69	-93	24	5	10	-68	-1	1	4
177	9	-73	-65	-8	4	6	-71	-2	2	4
178	10	-45	-44	-1	1	4	-44	-1	1	4
179	11	84	28	56	6	12	84	0	0	4
180	0	-73	70	-143	8	16	-71	-2	2	4
181	1	-50	-75	25	5	10	-54	4	3	5
182	2	59	81	-22	5	10	60	-1	1	4
183	3	-5	67	-72	7	14	-6	1	1	4
total bits						176	66			
bits per value						11.00	4.12			
compression ratio						1.455	3.879			

Table 23: The same window in polar coordinates — the mechanism. Against ramp $m - 1$ the phase step is unpredictable; against ramp $m - n_{\text{Tx}}$ it is essentially zero (a stationary target seen by the same antenna).

Quantity over ramps 168–183	vs. ramp $m - 1$	vs. ramp $m - n_{\text{Tx}}$
Magnitude change $\Delta x $, range	-21.8 to +20.3	-4.0 to +3.4
Phase step $\Delta\theta$, range (rad)	-2.263 to +2.622	-0.040 to +0.043
Phase step, mean $ \Delta\theta $ (rad)	1.104	0.018

7 Figures

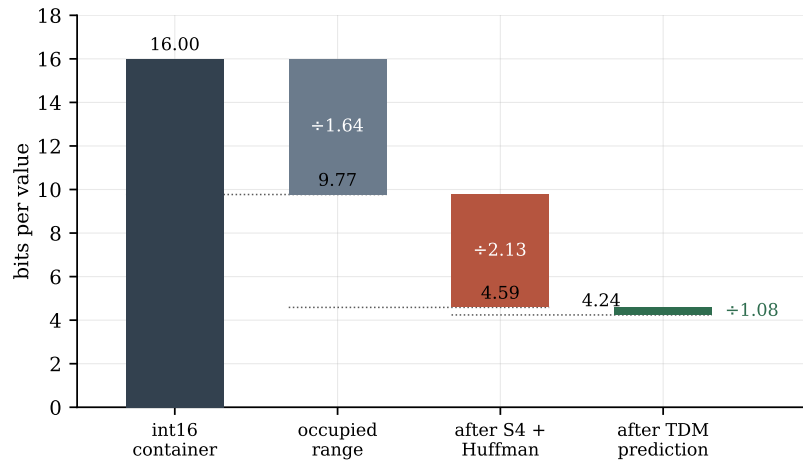


Figure 1: Decomposition of the compression ratio. Of the 16 bits in the `int16` container, 6.2 are structurally unused; the entropy coder recovers 2.13 from spectral sparsity; corrected prediction adds 1.08.

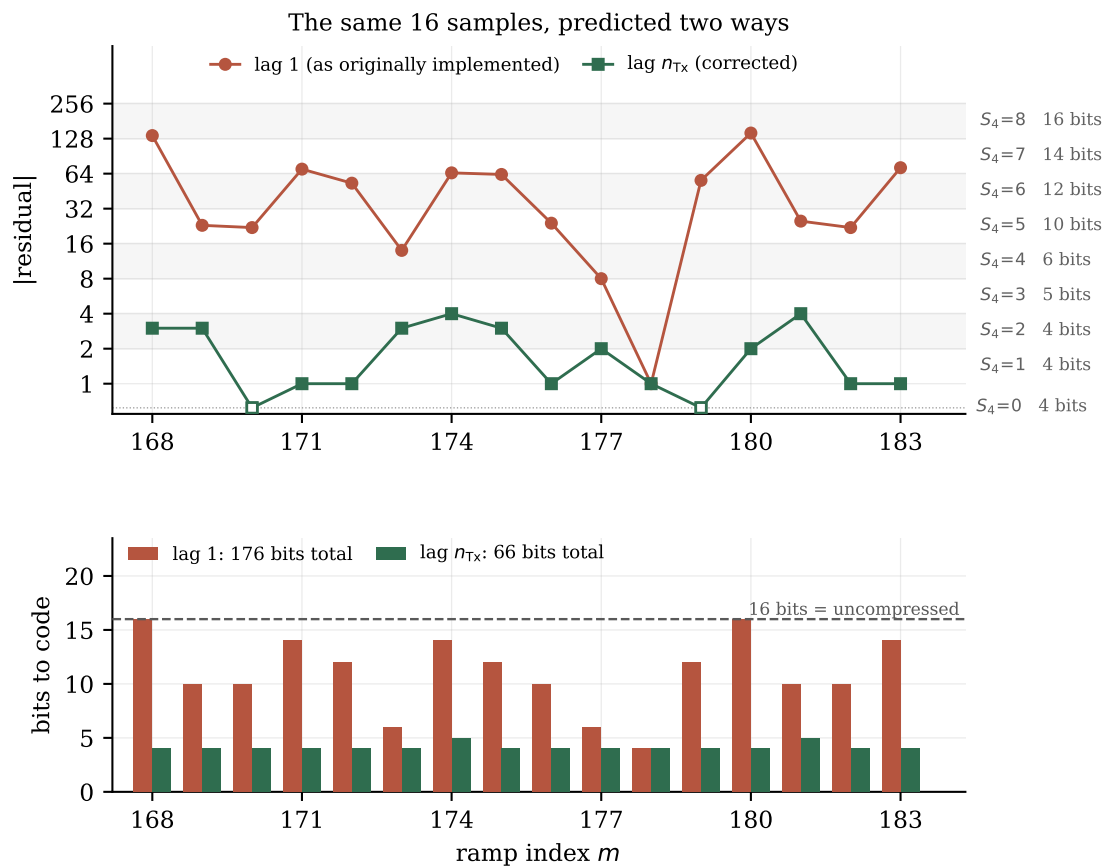


Figure 2: The 16-sample worked example plotted. Above: residual magnitude with the S_4 cost bands marked. Below: bits per sample against the 16 uncompressed. 176 bits versus 66, for identical information.

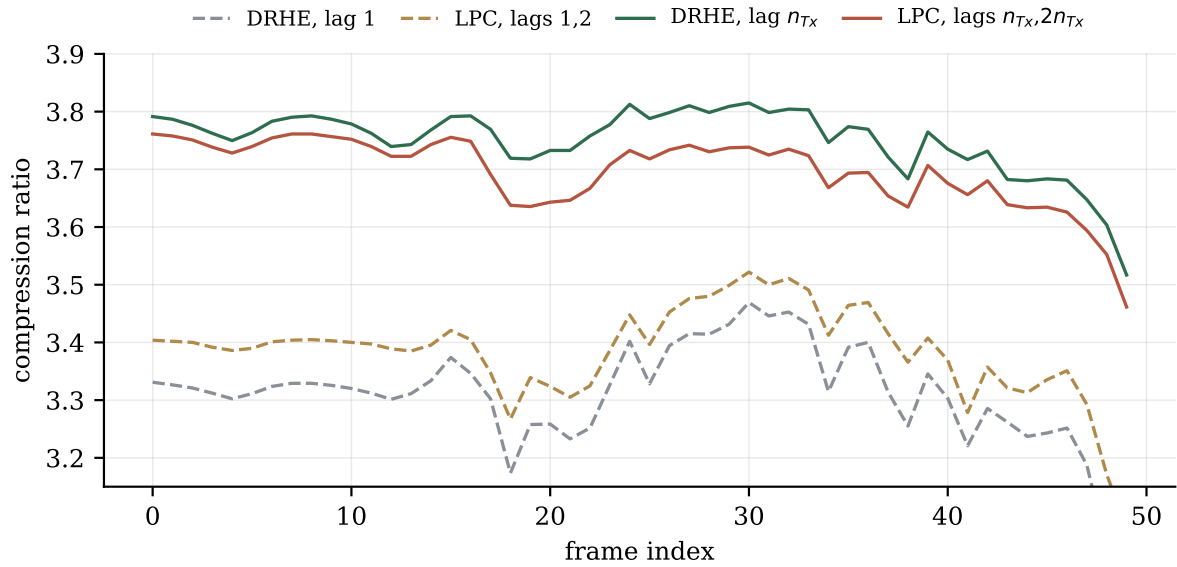


Figure 3: Per-frame compression ratio, all 50 frames. The corrected designs (solid) beat their baselines (dashed) on every frame, not just on average.

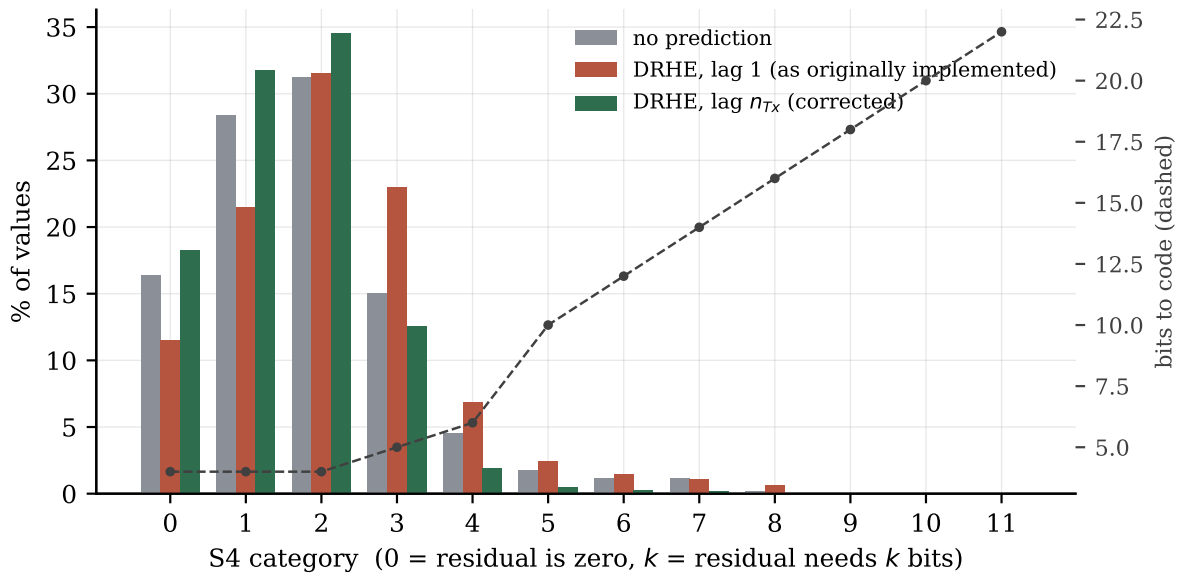


Figure 4: S4 category distribution. Lag-1 prediction pushes values *up* the cost scale relative to no prediction; lag- n_{Tx} pulls them back down.

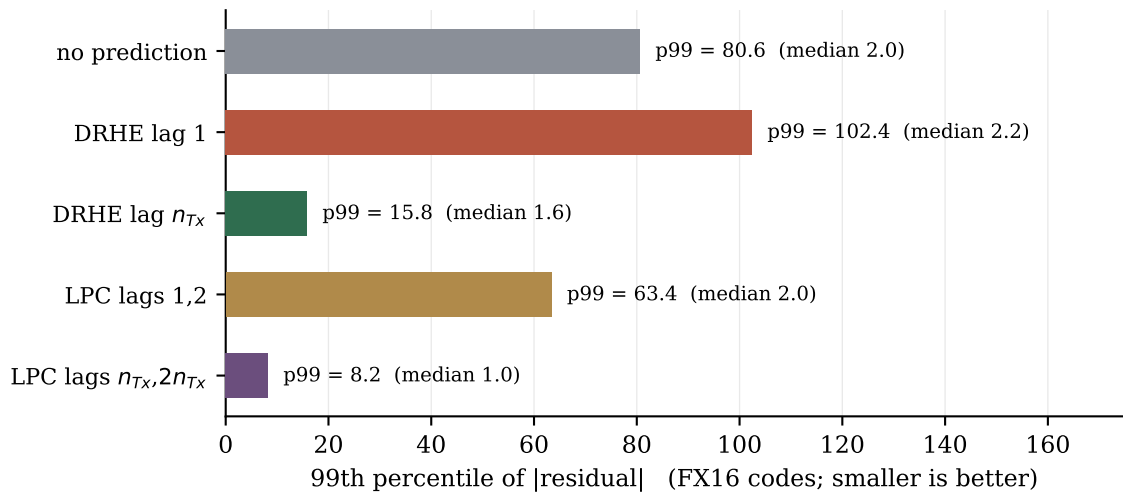


Figure 5: 99th percentile of residual magnitude — the clearest single view. Lag-1 DRHE has a *heavier* tail than no prediction at all.

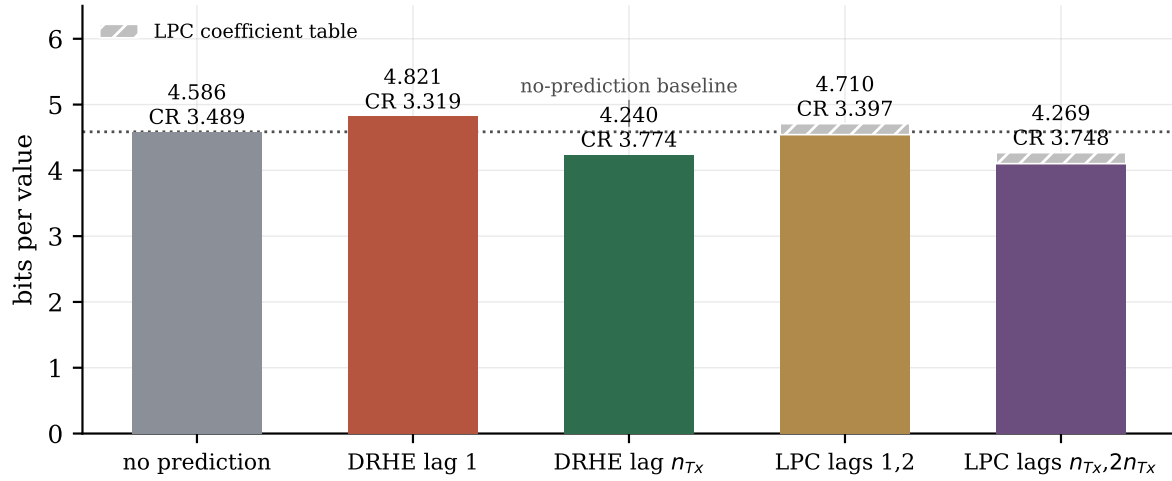


Figure 6: Bits per value, all five configurations. Dotted line is the no-prediction baseline: any bar above it is a predictor doing harm.

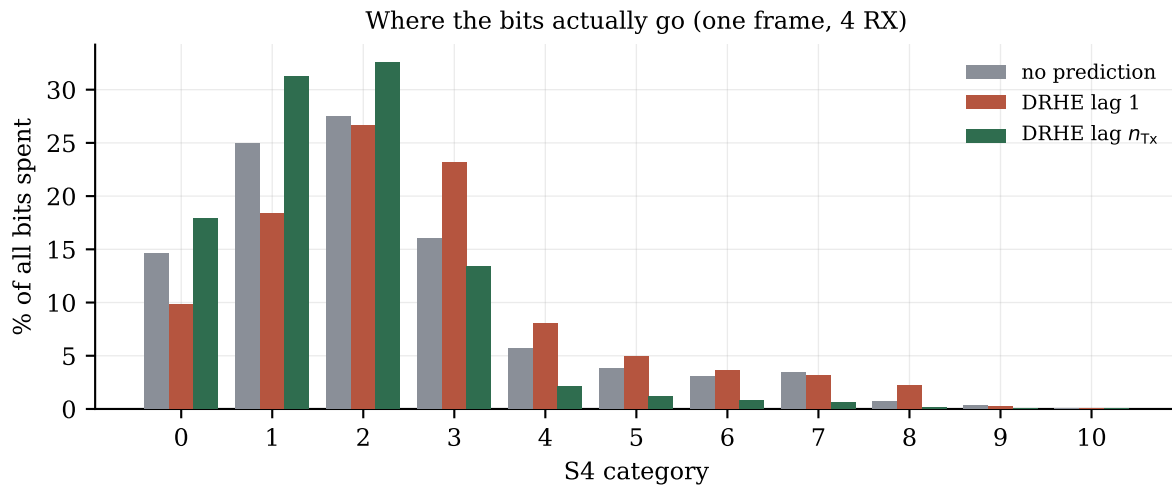


Figure 7: Share of total bits by S4 category. The correction concentrates spending into the three four-bit categories.

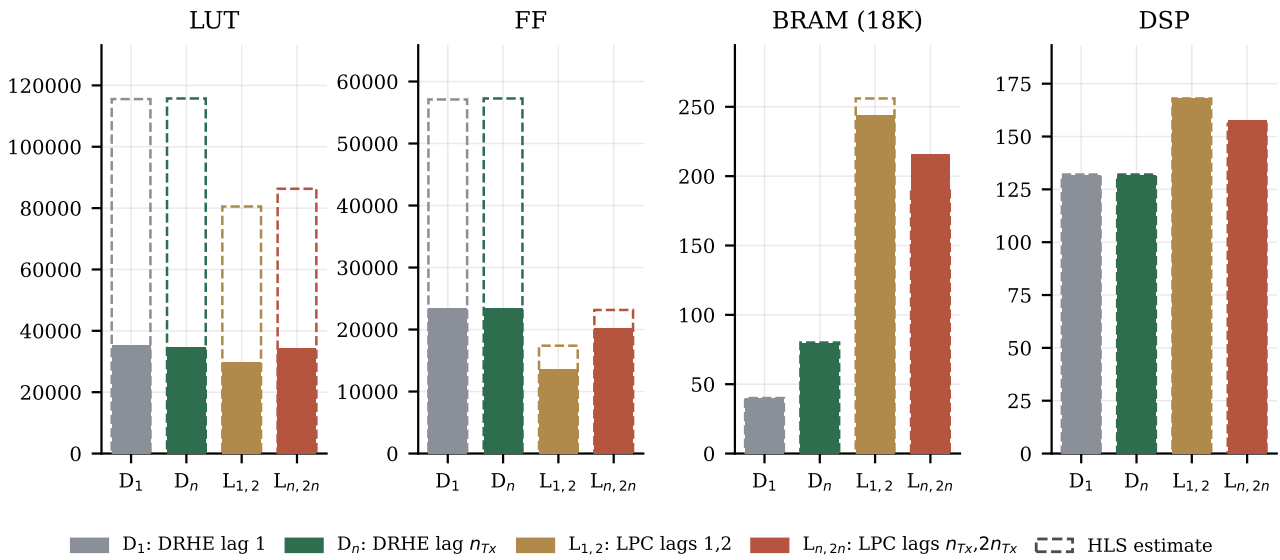


Figure 8: Resource usage, four designs, post-route (solid) against the HLS estimate (dashed outline). Implemented, the designs use similar logic and differ in memory.